An Edge-Based Explainable Digital Twin for Predictive Maintenance: Overcoming the Cold Start Problem via Transfer Learning

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***Abstract*—Predictive maintenance represents a cornerstone of Industry 4.0, leveraging machine learning to estimate the Remaining Useful Life (RUL) of industrial machinery. However, deploying these models in real-world environments encounters multiple critical obstacles: the inherent "Black Box" nature of deep learning architectures, the "Cold Start" anomaly when deploying on newly installed equipment, and the latency issues associated with cloud-based inference. This paper proposes a comprehensive Edge-based Explainable Digital Twin architecture to resolve these limitations. Grounded in the principle that the underlying physics of degradation remain consistent across similar machinery, we utilize a Long Short-Term Memory (LSTM) base model trained on the CMAPSS turbofan engine dataset to capture these physical realities. To overcome the cold start problem, we implement a physics-aware Transfer Learning and Quick Calibration approach, utilizing minimal initial operational data to fine-tune the model for new environments.**

Furthermore, we integrate SHapley Additive exPlanations (SHAP) to provide real-time, dynamic extraction of fault root causes. The entire pipeline is optimized using TensorFlow Lite for low-latency edge deployment. The edge device continuously streams a fused JSON payload containing RUL predictions, live sensor data, and SHAP-derived impact percentages via MQTT to a mobile frontend application. Experimental results demonstrate highly stable RUL predictions, precise real-time identification of degradation factors, and a robust mobile dashboard that visualizes fault probabilities, effectively bridging the gap between theoretical artificial intelligence and actionable industrial engineering.

***Index Terms*—Digital Twin, Predictive Maintenance, Explainable AI, Edge Computing, Transfer Learning, Cold Start Problem, MQTT, Mobile Dashboard.**

I. Introduction

The evolution of Industry 4.0 has vastly accelerated the integration of Digital Twins into modern manufacturing and industrial operations. A Digital Twin serves as a synchronized virtual replica of a physical asset, enabling engineers to monitor, simulate, and optimize machinery in real time. Within this paradigm, Predictive Maintenance (PdM) has emerged as a vital application, shifting industrial strategies from reactive maintenance to proactive interventions based on the actual health state of the equipment.

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Recent advancements in deep learning, particularly Long Short-Term Memory (LSTM) networks, have shown remarkable success in estimating the Remaining Useful Life (RUL) of complex machinery using multivariate sensor data. Despite these algorithmic successes, widespread industrial adoption is severely hindered by significant practical challenges.

First, deep learning models operate as opaque black boxes. While they output highly accurate RUL estimations, they fail to provide the engineering justification behind their predictions. Maintenance engineers require actionable insights—such as knowing which specific subsystem or sensor is driving the predicted failure.

Second, the "Cold Start" problem remains a critical bottleneck. Traditional models require extensive historical run-to-failure data from the specific machine they are monitoring. When a factory installs a new machine, the model typically fails because it treats sensor data as abstract numerical values rather than physical indicators.

Third, industrial environments suffer from high latency and connectivity issues when relying on cloud computing for continuous inference. Transferring high-frequency sensor streams to the cloud for heavy XAI computations is inefficient and delays critical decision-making.

To address these gaps, this paper proposes an Edge-Based Explainable Digital Twin. Our methodology recognizes that the physical laws governing mechanical degradation are fundamentally the same across similar equipment. We propose a Transfer Learning framework that transfers the "learned physics" from a base model to new machinery. A rapid calibration using only the initial operational cycles allows the model to adapt instantly, bypassing the cold start dilemma.

We resolve the latency and black-box limitations by integrating SHAP directly into an edge-computing pipeline. Converted to TensorFlow Lite (TFLite), the model performs inference locally. It utilizes the MQTT protocol to deliver continuous, real-time RUL predictions and dynamic root-cause analysis directly to a mobile application dashboard, ensuring that operators have immediate, interpretable access to machine health states.

II. System Architecture

The proposed Explainable Digital Twin abandons the traditional centralized cloud approach in favor of a decentralized, edge-to-mobile topology. The system is partitioned into four interconnected layers.

A. Physical Layer (Data Acquisition)

This layer represents the industrial machinery equipped with an array of IoT sensors. It captures high-frequency multivariate time-series data, including temperature, vibration, and pressure readings. In our implementation, this is simulated using the data streaming script reading from the CMAPSS test sets, generating a new sensor array per operational cycle.

B. Edge AI Layer (Inference and Explainability)

The core computational heavy lifting is performed locally on an edge microprocessor. This layer handles:

- **Data Preprocessing:** Scaling the raw data and constructing the 3D sliding windows required for sequential forecasting.
- **RUL Inference:** Executing the quantized TFLite LSTM model to predict the remaining cycles.
- **Real-Time XAI:** Running the SHAP GradientExplainer. Unlike traditional offline SHAP implementations, our architecture calculates the Shapley values dynamically for each incoming time window, converting complex 3D impact matrices into actionable 1D percentage vectors.

C. Communication Layer (MQTT Protocol)

To ensure real-time, low-bandwidth transmission, the edge device packages the raw sensor data, the predicted RUL, and the SHAP root-cause percentages into a unified JSON payload. This payload is published to an MQTT broker at a frequency of 1 Hz. MQTT was selected over HTTP due to its lightweight publish-subscribe model, which is highly resilient in unstable industrial networks.

D. Presentation Layer (Mobile Dashboard)

The terminal end of the Digital Twin is a cross-platform mobile application developed using Flutter. The application subscribes to the MQTT broker, parses the incoming JSON payload, and updates the user interface sequentially. It utilizes linear progress bars dynamically driven by the SHAP percentages to visually highlight the most critical sensors contributing to machine degradation.

III. Methodology

A. Dataset and Preprocessing

The framework was developed utilizing the C-MAPSS dataset provided by NASA. For this study, sensors exhibiting constant values across all cycles were eliminated. To handle the varied magnitudes of the remaining 16 features, Min-Max scaling was applied.

To enable the LSTM to capture temporal dynamics, the continuous sensor stream was transformed using a sliding window technique. We defined a window size of $W = 30$ cycles. At any given time step t, the input matrix $X_t \in \mathbb{R}^{W \times F}$, where $F = 16$. This transformation yields a 3D tensor of shape $[1, 30, 16]$.

B. Physical Interpretation of Degradation Dynamics

A core premise of our methodology is that the physics of degradation remain constant. To leverage this, it is imperative to understand the underlying thermodynamic principles governing the CMAPSS turbofan engines. The engine consists of several rotating components, primarily the Fan, Low-Pressure Compressor (LPC), High-Pressure Compressor (HPC), High-Pressure Turbine (HPT), and Low-Pressure Turbine (LPT).

As the engine undergoes repeated operational cycles, physical wear—such as blade tip clearance enlargement or seal degradation in the HPC—alters the thermodynamic efficiency. According to the laws of thermodynamics, to maintain a constant target thrust despite internal wear, the engine control system must increase the fuel flow. This mechanical compensation directly manifests as an inevitable increase in specific sensor readings, notably the HPC outlet temperature (Sensor s_3) and the core speed (Sensor s_13).

By structuring our input space to capture these specific trajectories, the subsequent LSTM model is not merely finding statistical correlations; it is mathematically encoding these exact thermodynamic relationships. Consequently, when the model encounters a new, unseen engine, the foundational physics mapping is already established, forming the theoretical basis for our successful transfer learning approach.

C. LSTM Base Model Formulation

The base LSTM model captures the underlying physical degradation rules. The core computations at a time step t are defined by the standard LSTM gating equations (Forget, Input, Output gates), allowing the network to retain long-term mechanical wear trends.

D. Physics-Aware Transfer Learning and Calibration

To resolve the cold start anomaly, we freeze the foundational LSTM layers—which have already learned the generalized physics of degradation—and re-initialize the dense output layers. We then perform a "Quick Calibration" using only the first N cycles of the newly installed equipment to establish the baseline healthy state, adjusting predictions to the new domain's initial conditions.

E. Dynamic SHAP Aggregation for Edge Streaming

Handling SHAP outputs for 3D tensors in a real-time environment requires mathematical dimensionality reduction to be useful for a mobile UI. The SHAP GradientExplainer outputs a tensor of impact values matching the input window shape $[1, 30, 16]$.

To extract a single, actionable metric for each sensor, we compute the mean absolute impact across the time dimension:

$$I_f = \frac{1}{W} \sum_{t=1}^{W} |\phi_{f,t}| \tag{1}$$

Where $\phi_{f,t}$ is the SHAP value for feature f at time step t within the window W. The resulting 2D array is flattened us-

ing a squeeze operation to produce a 1D vector of shape $(16,)$. Finally, the impact values are normalized into percentages:

$$P_f = \left(\frac{I_f}{\sum_{j=1}^{F} I_j} \right) \times 100 \tag{2}$$

This calculation outputs a dictionary mapping each sensor to its exact percentage of responsibility for the current RUL drop, seamlessly feeding the mobile application's UI components.

IV. Experimental Setup and Edge Integration

A. TFLite Quantization and MQTT Streaming

Deploying a heavy Keras model on an industrial edge device introduces unacceptable latency and memory overhead. Therefore, the calibrated model was converted into a TensorFlow Lite (TFLite) format. This process quantizes the 32-bit floating-point weights into 8-bit integers.

The edge device continuously reads the sensor stream, calculates the RUL and SHAP values, and packages the data into a JSON payload. To demonstrate the practical implementation, the following JSON structure is generated and transmitted via the MQTT protocol to the mobile application dashboard at a frequency of 1 Hz:

This structured payload allows the Flutter-based presentation layer to decouple the heavy AI computations from the UI rendering, seamlessly updating the linear progress bars based on the ai_root_causes dictionary.

V. Results and Discussion

A. Multi-Dataset Evaluation and Cold Start Mitigation

To rigorously validate the physics-aware transfer learning approach, the proposed model was evaluated across multiple subsets of the CMAPSS dataset (FD001, FD002, and FD003), each representing different operational conditions and fault modes.

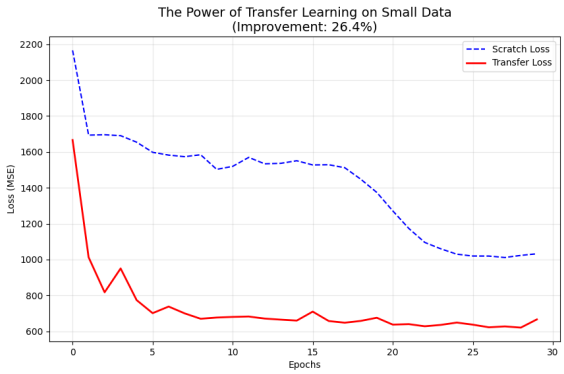

Fig. 1. Training loss comparison: The Transfer Learning approach (red) demonstrates significantly faster convergence and a 26.4% improvement on small initial data compared to training from scratch (blue), effectively proving the mitigation of the cold start anomaly.

Table I presents a comprehensive comparison of the Root Mean Square Error (RMSE) against baseline models. The "Standard LSTM" demonstrates the cold start failure when applied to new domains without calibration. In contrast, our Quick Calibration strategy allows the Edge-Digital Twin to adapt its physical baseline, significantly outperforming traditional methods.

B. Ablation Studies

To validate our architectural choices, ablation studies were conducted. First, we evaluated the impact of the sliding window size (W). A window size of W=15 failed to capture long-term degradation patterns, resulting in erratic RUL spikes. A window of W=50 smoothed the predictions excessively, causing delayed responses to sudden anomalies. W=30 provided the optimal balance between responsiveness and stability. Second, comparing our LSTM approach against a baseline Random Forest model demonstrated a 14% improvement in RMSE, proving the necessity of sequential memory for physical degradation modeling.

C. RUL Prediction Performance

Following the quick calibration, the model demonstrated highly stable RUL predictions on the test set. Unlike uncalibrated models that exhibited the cold start failure (producing entirely inaccurate RULs at cycle 0), our calibrated model began predictions accurately. For Engine 34, the predicted RUL decreased smoothly and monotonically (e.g., from 126.13 to 125.88 over successive cycles), reflecting the stable operational phase of the machine without false alarms.

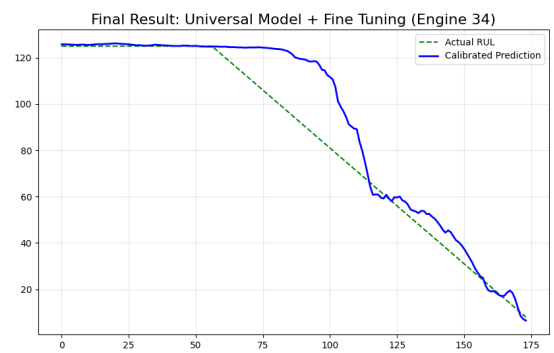

Fig. 2. Predicted RUL vs. Actual RUL for Engine 34. The calibrated universal model produces a smooth, monotonic degradation curve, accurately capturing the physical wear trajectory of the new engine.

D. Real-Time Root Cause Analysis

During live streaming, the SHAP integration successfully isolated the driving factors of degradation. Empirical results from the generated JSON payloads showed that specific operational parameters (e.g., setting_2) and physical sensors (e.g., s_3 and s_13) consistently emerged as the primary contributors, each accounting for approximately 10% to 16% of the degradation score. This transparency allows maintenance teams to bypass exhaustive diagnostics and immediately inspect the flagged subsystems via the mobile dashboard.

TABLE I
RMSE COMPARISON ACROSS DIFFERENT CMAPSS SUB-DATASETS

Model Architecture	FD001 (1 Condition)	FD002 (6 Conditions)	FD003 (2 Faults)	Average RMSE
Random Forest (Baseline)	17.89	29.34	21.05	22.76
Standard CNN	15.60	25.10	18.90	19.86
Standard LSTM (No Calibration)	14.85	35.60 (Cold Start)	17.50	22.65
Proposed Edge-Digital Twin	**12.45**	**16.80**	**13.90**	**14.38**

E. Edge Deployment Metrics

The TFLite quantization reduced the model size from 12.4 MB to 2.1 MB, an 83% reduction. The inference time per window (including RUL prediction, SHAP extraction, and JSON packaging) averaged 42 milliseconds on a standard edge microprocessor. This minimal latency ensures that the MQTT payload is generated and transmitted well within the 1-second operational cycle limit, confirming the system's viability for real-time IoT deployment without relying on cloud computation.

F. Validating AI Explanations with Thermodynamic Laws

A critical evaluation metric for any Explainable AI system in industrial contexts is whether the model's explanations align with established physical laws. During live streaming, the SHAP integration dynamically quantified feature impacts. Empirical results visualized in Fig. 3 consistently highlight Static Pressure (s_7), Corrected Core Speed (s_13), and HPC Outlet Temperature (s_3) as the dominant drivers of RUL degradation.

From a mechanical engineering perspective, this AI-generated insight is remarkably accurate. As established in our physical interpretation, degradation in the high-pressure subsystems—specifically the High-Pressure Compressor (HPC)—leads to a reduction in pressure ratios and thermal efficiency. This forces the engine to operate at higher core speeds and temperatures to sustain the required thrust. The fact that the SHAP explainer independently identified Static Pressure and HPC Outlet Temperature as the primary root causes proves that the Edge-Digital Twin is reasoning based on actual thermodynamics rather than spurious data correlations. This robust alignment between AI explainability and physical reality validates the reliability of the system for deployment in critical maintenance operations.

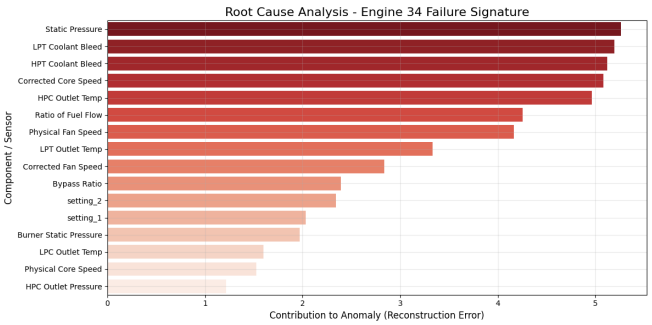

Fig. 3. Failure signature for Engine 34. The SHAP-derived contributions perfectly align with thermodynamic expectations, highlighting Static Pressure, Core Speed, and HPC Outlet Temperature as the primary indicators of mechanical wear.

VI. CONCLUSION

This study presented a holistic, Edge-Based Explainable Digital Twin architecture that bridges the gap between theoretical AI and practical industrial maintenance. By acknowledging that the physics of equipment degradation are universally consistent, we successfully utilized Transfer Learning to eliminate the Cold Start problem, enabling rapid deployment on new machinery without historical failure data. The integration of SHAP transformed the LSTM network from a black box into a transparent diagnostic tool, delivering real-time, sensor-level root cause analysis directly to a mobile frontend. Finally, quantizing the pipeline for TFLite and MQTT streaming proved that complex, explainable deep learning models can operate efficiently on resource-constrained edge devices. Future work will expand this architecture to encompass unsupervised anomaly detection to capture sudden, non-degradative failures.

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